

Juan Antonio G. Soria Rodríguez

Software & AI/ML Engineer | NLP · LLMs · Full-Stack

📍 Mexico City, Mexico (GMT-6) ✉ antonio.soria8r@gmail.com

🟡 PurpleAlien 📺 [antonio-sould](https://antonio-sould.github.io)

🌐 purplealien.github.io



Professional Summary

Final-year Computer Science student at Universidad Autónoma Metropolitana (UAM-I, 100% of credits expected Dec 2026, degree in progress), specialized in Artificial Intelligence, Natural Language Processing (NLP), and AI-augmented software development. Completed a research-grade Bachelor's thesis building a production-level, dual-mode paraphrase and plagiarism detection system over an institutional corpus of ~5,000 theses — combining deep learning, vector retrieval, transformer fine-tuning, and containerized web deployment.

Completed the coursework for a first degree in Applied Mathematics and Computing (UNAM); then spent several years working, building personal projects and self-studying before formalizing this focus with the B.Sc. in Computer Science (UAM-I) in late 2020.

Actively uses AI-augmented methodologies: context engineering, agentic workflows, AI-assisted code generation, and multi-agent system design. Views AI as a strategic multiplier for analyzing complex problems, validating solutions, and transforming ideas into functional products. Seeking to join a remote team as an ML/NLP or Backend Engineer, taking AI models and systems to production.

Education

B.Sc. in Computer Science

2021 – 2026

[Universidad Autónoma Metropolitana \(UAM-I\), Mexico City](#)

Thesis (Capstone Projects I & II — 100% of credits: Dec 2026, degree in progress): Intelligent Paraphrase Detection using Large Language Models (LLMs)

Advisor: Dr. Benjamín Moreno Montiel

Capstone II (in progress): sentiment analysis of the Codex Gigas — NLP for Latin with contextual language models

Applied Mathematics and Computing

2011 – 2016

[Universidad Nacional Autónoma de México \(UNAM\), FES Acatlán](#)

Coursework completed; strong mathematical foundation (calculus, linear algebra, probability and statistics) underpinning my AI/ML work. Degree completion in progress.

Capstone Project I: Paraphrase & Plagiarism Detection System

End-to-end, dual-mode NLP system (pairwise classification + corpus search) for academic plagiarism detection at UAM-I. Deployed over ~5,000 theses with a FAISS semantic index of **2.28M vectors at 100% corpus coverage**.

- **Architecture:** Bi-encoder MiniLM + FAISS → cross-encoder XLM-RoBERTa/BETO → optional LLM arbiter (Llama 3.3 70B) in the decision boundary [0.35, 0.65].
- **Models:** Fine-tuned XLM-RoBERTa (calibrated, T=5.15, ECE 0.076), BETO v8a (F1=0.889), soft-voting ensemble with **AUROC 0.999**. Rigorous evaluation with bootstrapped confidence intervals and data-leakage audit.
- **Key finding:** The CBI corpus is ~97% original (only ~3% real paraphrase). Core contribution is data efficiency: **60 human pairs match 180K algorithmic pairs**.
- **Delivery:** Unified web app with Docker, PWA, section-level similarity reports, heatmaps, and LLM-powered explainability.

Stack: Python, PyTorch, Hugging Face Transformers, sentence-transformers, FAISS, MongoDB, Docker.

More details: purplealien.github.io/proyecto-terminal.html

Technical Skills

Languages: Python (Advanced) Java (Experienced) C / C++ JavaScript PHP

ML / NLP: PyTorch Hugging Face BERT / XLM-R sentence-transformers FAISS LLMs / RAG scikit-learn

AI-Augmented: Prompt Engineering Context Engineering Agentic AI AI-Assisted Code Multi-Agent

Tools: Git / GitHub Docker AWS (basic) MongoDB MySQL Linux Jupyter / Colab

Web / Backend: Spring Boot Next.js / React FastAPI Node.js PostgreSQL REST / Microservices

Professional Experience

Freelance Software Developer 2022 – Present
[Independent](#)

Custom web systems for clinics and small businesses — management, scheduling and invoicing — built with Spring Boot, Next.js and PostgreSQL, deployed to production.

Adjunct Instructor (freelance) Feb 2020 – Dec 2021
[Admission and remedial courses](#)

High-school and university admission courses plus Algebra, Geometry, Calculus and Physics; designed course content and adapted it to each group.

IT Support Technician Apr 2017 – Apr 2021
[BTL Company](#)

Computer maintenance, software and OS installation, and day-to-day user support.

Security Systems Installer Jul 2016 – Aug 2019
[SYSARK](#)

Installation and configuration of CCTV and access-control systems; on-site project coordination and delivery.

Highlighted Personal Projects

Full-Stack Management Systems

End-to-end systems for veterinary and dental clinics: **Spring Boot 3** + PostgreSQL + JWT + WebSockets (interactive odontogram, PDF invoicing) and a **Next.js 15** + Turborepo + Drizzle ORM monorepo deployed on Vercel.

Microservices Architecture

E-commerce and pharmacy platforms with up to **8 Spring Boot microservices**, RabbitMQ, API Gateway, and Docker Compose orchestration.

Artificial Intelligence & Algorithms (Java)

Portfolio of 15+ projects: BFS, DFS, A*, Minimax, Negamax, and Alpha-Beta pruning. Game logic for Connect 4, Tic-Tac-Toe, N-Queens, and sliding puzzles using Java Swing.

Low-Level Systems & Compilers (C)

Spanish→C pseudocode compiler with **Flex & Bison** (lexical, syntactic, semantic analysis, and code generation). Operating Systems and Concurrent Programming projects in C (processes, threads, shared memory).

Game Development

2D Mario-style platformer in Java (Swing, MVC architecture, Gradle) and a low-poly 3D open world in **Unity / C#**.

Relevant Coursework

Linear Algebra | Calculus | Discrete Mathematics | Probability & Statistics | Software Engineering | Operating Systems | AI | Machine Learning | Databases | Discrete Simulation | Networks | Algorithm Design & Analysis

Languages: Spanish (Native) | English (Intermediate / Technical reading & writing) | Portuguese (Basic)

Mexican nationality. Available for remote ML/NLP Engineer and Backend (Python/Java) roles, junior to mid level. Immediately available, full-time, fully remote.